Matlab Set 4

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Exercise: Given the surface charge density, $\rho_s = 2.0\ \mu C/m^2$, existing in the region $\rho < 1.0$ m, $z = 0$, and zero elsewhere, find **E** at $P(\rho = 0, z = 1.0)$ and write a MATLAB program to verify your answer.

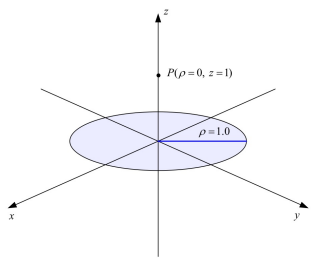


Figure 4.6 The problem of the exercise of Set 4.

- $\rho_s = 2\mu\ C/m^2$
- in region:
  - $\rho < 1m$
  - $z = 0$
- $P(\rho = 0, z = 1) \rightarrow P(0, 0, 1)$
- $\mathcal{E}$ @ $P = ?$

- $d\vec{\mathcal{E}} = \frac{dQ}{4\pi\varepsilon_0 R^2}\vec{a_r}$

$= \int_S \frac{\rho_s dS}{4\pi\varepsilon_0 R^2}\vec{a_r}$

$= \iint \frac{\rho_s \rho\, d\phi\, d\rho\, \vec{R}}{4\pi\varepsilon_0 |R|^3}$

$= \iint \frac{\rho_s \rho \cdot (\rho(-\vec{a_\rho}) + z\vec{a_z})}{4\pi\varepsilon_0 \sqrt{\rho^2 + z^2}^3}\, d\phi\, d\rho$

$= \frac{\rho_s}{4\pi\varepsilon_0}\int_{\phi=0}^{2\pi}\int_{\rho=0}^{1} \frac{\rho z\ \vec{a_z}}{\sqrt{\rho^2 + z^2}^3}\, d\phi\, d\rho$

$= \frac{\rho_s z}{4\pi\varepsilon_0} 2\pi \int_{\rho=0}^{1} (\rho^2 + z^2)^{-3/2}\ \frac{1}{2}\ d(\rho^2)\ \vec{a_z}$

$= \frac{\rho_s z}{2\varepsilon_0}\left(-[\rho^2 + z^2]^{-1/2}\right)\Big|_0^{\rho=0}\vec{a_z}, \quad z = 1 - 0 = 1$

$= \frac{\rho_s(1)}{2\varepsilon_0} -(\rho^2 + 1^2)^{-1/2}\Big|_0^1\ \vec{a_z}$

$= \frac{\rho_s}{2\varepsilon_0}\left(-[(1)^2 + 1]^{-1/2} - -(0^2 + 1)^{-1/2}\right)\vec{a_z}$

$= \frac{\rho_s}{2\varepsilon_0}\left(-(2)^{-1/2} + 1^{-1/2}\right)\vec{a_z}$

$= \frac{\rho_s}{2\varepsilon_0}\left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{1}}\right)\vec{a_z}$

$= \frac{\rho_s}{2\varepsilon_0}\left(1 - \frac{1}{\sqrt{2}}\right)\vec{a_z}$

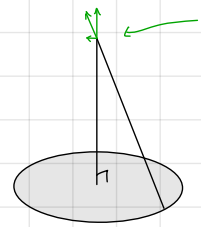
$= \frac{2\mu}{2\varepsilon_0}\left(1 - \frac{1}{\sqrt{2}}\right)\vec{a_z}$

$= 33079\ \vec{a_z}$

$\doteq 33080\ \vec{a_z}$

$\vec{\mathcal{E}} = 3.308 \times 10^4\ m\ \vec{a_z}$

- $R = \rho(-\vec{a_\rho}) + z\vec{a_z}$
- $|R| = \sqrt{\rho^2 + z^2}$
- $\vec{a_R} = \frac{\vec{R}}{|R|}$
- $dQ = \rho_s dS = \rho_s \rho\, d\phi\, d\rho$



$\vec{\mathcal{E}_x}$ & $\vec{\mathcal{E}_y}$ cancel out $\rightarrow 0 \le \phi \le 2\pi$
only $\vec{\mathcal{E}_z}$ remains.